Dr.Syed Mohsin Ali Shah

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Professional Summary

ML Engineer and researcher with a Ph.D. degree in Engineering and 6+ years of experience across industry and academia, specializing in the design and deployment of production-grade AI systems. I build and operate end-to-end data and model pipelines, including dataset creation and validation, training, fine-tuning, evaluation, and scalable deployment. My work is primarily in Python using PyTorch/TensorFlow, FastAPI, and Docker across CPU, GPU, and distributed/HPC environments. I develop Agentic AI and LLM-powered systems, including Retrieval-Augmented Generation (RAG) pipelines built with embeddings and vector databases for enterprise search and decision intelligence. I design structured evaluation frameworks with statistical validation and continuous monitoring, and apply post-training improvements such as supervised fine-tuning and preference-based optimization, along with inference optimization and model compression to meet latency and scalability targets. I combine strong research foundations with practical engineering to deliver reliable, maintainable, and high-performance AI solutions.

Education

University of Deusto **Bilbao, Spain**
Ph.D. in Engineering for Information Society and Sustainable Development *2023 – 2026*
Research focused on the development of machine learning and AI-driven image processing methods for low-cost environmental pollution monitoring, designed for scalable and data-driven citizen science applications.

Bahria University **Islamabad, Pakistan**
M.S. in Computer Engineering *2019 – 2021*
GPA: 3.8/4.0; Thesis: Consumer Emotion Prediction Using Machine Learning.

Bahria University **Islamabad, Pakistan**
B.S. in Computer Engineering *2013 – 2017*

Professional Experience (6+ Years)

Multiverse Computing **Zaragoza,Spain**
Machine Learning Engineer *Feb 2026 – Present*

- Build end-to-end data and model pipelines, including dataset creation, sourcing, augmentation, validation, and automated training/fine-tuning/evaluation workflows.
- Develop and deploy models that meet product and customer requirements, ensuring reliability and measurable performance improvements.
- Prototype and experiment with foundational LLM techniques, evaluating novel architectures and training methods to push performance boundaries in NLP and generative AI contexts.
- Implement model compression and optimization techniques to meet performance, scalability, and deployment constraints.
- Collaborate closely with ML engineers, researchers, and software engineers on data curation, evaluation design, model serving, and observability.
- Contribute clean, well-tested Python code to shared repositories, document design decisions, and maintain high engineering standards.

QUEBEC INC, **Remote, Ontario,Canada**
Machine Learning Engineer *Nov 2025 – Jan 2026*

- Designed, trained, and deployed production-grade Machine Learning, Deep Learning, and LLM-based systems supporting arbitrage, user acquisition, and marketing automation pipelines.
- Developed end-to-end ML workflows including data ingestion, feature engineering, model training, evaluation, deployment, and continuous monitoring.
- Collaborated closely with Data Analysts to translate insights into deployable ML solutions, forming the foundation of a data-driven growth strategy.
- Evaluated and optimized model performance using statistical analysis, A/B testing, and continuous feedback loops.

Aurakore IT Solutions **Remote, Austin, TX, USA**
Machine Learning Engineer *Mar 2025 – Oct 2025*

- Designed and maintained multi-agent AI systems based on the Model Context Protocol (MCP), supporting coordinated task execution for memory tracking, outreach automation, and real-time entity discovery.
- Developed and deployed FastAPI-based microservices to enable secure and modular communication between AI agents across cloud and on-premise infrastructures.
- Implemented structured memory agents to persist contextual histories, enabling personalized interaction management and long-term behavioral adaptation.
- Automated data processing and quality assurance workflows, including consistency checks, fallback resolution strategies, and prompt validation layers for safe inference.
- Integrated real-time data pipelines into agent logic using containerized deployments and CI/CD pipelines, ensuring reproducibility, isolation, and minimal downtime.
- Created internal documentation and technical manuals for agent orchestration, deployment procedures, and module-specific configurations to support onboarding and maintenance.
- Fine-tuned domain-specific LLMs with custom datasets, applying low-rank adaptation (LoRA) and quantization for resource-efficient inference.
- Evaluated LLM performance using NLP benchmarks (MMLU, TruthfulQA) and created business-specific evaluation sets to align outputs with operational goals.

DeustoTech, University of Deusto **Bilbao, Spain**

Research Assistant *Mar 2023 – Feb 2025*

- Designed and deployed machine learning pipelines for real-time particulate matter estimation using low-cost image sensors, supporting urban air quality monitoring in resource-constrained environments.
- Developed automated workflows for data ingestion, preprocessing, and validation using Python (pandas, scikit-learn, TensorFlow), ensuring consistency and reproducibility in large-scale environmental datasets.
- Implemented quality control mechanisms including threshold filtering, anomaly detection, and data completeness checks to maintain the integrity of sensor streams.
- Maintained structured archives of PM sensor data and supported FAIR principles in dataset curation and versioning for long-term research accessibility.
- Built and maintained scalable RESTful APIs and backend services using FastAPI, enabling real-time access to harmonized data streams and model inference endpoints.
- Deployed containerized AI models on HPC infrastructure using Docker and GitHub Actions to automate training, testing, and version control workflows.
- Developed ML pipelines for PM10 estimation from low-cost image sensors; implemented preprocessing (particle detection + estimation experiments), and calibration against reference stations; deployed as containerized inference services (Docker + FastAPI) and validated against colocated reference stations.
- Designed and operationalized quality-control and anomaly-detection routines (threshold filters, temporal consistency checks) and a FAIR dataset curation strategy for long-term urban sensor archives used by citizen science campaigns.
- Integrated crowdsourced sensor data with geospatial information to conduct data assimilation experiments, applying time-series machine learning and image recognition models for high-resolution, city-scale exposure mapping.
- Authored peer-reviewed publications and presented applied research at international conferences in environmental AI and low-cost sensing.
- Created user documentation, data dictionaries, and technical guides to support collaborators and citizen science participants accessing and interpreting platform data. Project link: amiaire.org

Norwegian Institute of Air Research (NILU) **Oslo, Norway**

Visiting Researcher *Nov 2024 – Feb 2025*

- Conducted advanced research on the integration of image processing pipelines for environmental data analysis, focusing on preprocessing, enhancement, and compression of image samples.
- Performed comparative analysis of image processing-based estimation of particulate matter (PM) with actual measurements from NILU-operated air quality monitoring base stations.
- Designed and implemented a web-based platform utilizing image processing techniques, enabling citizen scientists to participate in data collection for air pollution monitoring.
- Developed and deployed machine learning models to automate PM detection and quantification, establishing robust validation protocols to ensure model reliability against reference-grade instrumentation.

SZABIST **Islamabad, Pakistan**

Jr. Lecturer & Lab Demonstrator *Jan 2019 – Feb 2023*

- Delivered undergraduate courses in Software Engineering, Deep Learning, Computer Vision, and Artificial Intelligence with hands-on labs and real-world case studies.
- Mentored students on industry-focused projects involving time-series forecasting, recommendation systems and voice recognition systems.
- Supervised projects on speech emotion recognition, voice activity detection, and automatic speech recognition (ASR) using deep learning frameworks.
- Taught core Software Engineering concepts including Software Development Life Cycle (SDLC), requirement analysis, system modeling, and use case diagram development.
- Assisted students in preparing software project documentation, including Software Requirement Specifications (SRS), UML diagrams, and technical reports.

Projects

AI-Powered Web Platform for Low Cost Image Processing Based Air Quality Monitoring:
- Architected a modular microservices-based platform for air pollution inference using containerized machine learning models in a hybrid edge–cloud environment. Amiaire Web App Link
- Developed RESTful APIs for seamless integration between image processing models, environmental sensor data services, and frontend visualization components.
- Built a scalable web application enabling users to visualize and interpret localized air quality trends through dynamic charts and maps.
- Adopted by over 10,000 users from more than 140 schools across Spain as part of environmental science education campaigns and outreach programs.
- Implemented a contributor dashboard allowing users to monitor and analyze their sensor-based data submissions, promoting transparency and public participation. Amiaire Map Contributions Link By Citizen Scientists

Anomaly Detection in Air Pollution Patterns Using Unsupervised Learning:
- Designed unsupervised learning workflows to identify deviations in pollution level distributions across monitoring locations.
- Conducted statistical and pattern-based analysis on structured air quality datasets to support anomaly classification and automated reporting.
- Optimized data preprocessing pipelines to handle missing values, noise reduction, and normalization for high-frequency air quality sensor data.
- Evaluated model performance using clustering metrics and anomaly scoring methods to refine detection accuracy and reduce false positives.

Air Quality Data Visualization and Monitoring:
- Developed a Python-based system for retrieving real-time air quality data (PM_{10}, NO_2, O_3) from municipal monitoring stations in Bilbao, Spain, via public environmental APIs.
- Designed automated routines to preprocess incoming data streams, including missing value handling, unit normalization, and timestamp alignment.
- Built interactive visualizations using Plotly and Matplotlib to display pollutant levels, daily averages, threshold exceedances, and historical trends.
- Created exportable reports and dashboard views to support scientific review, public transparency, and policy outreach.
- Implemented data archiving and versioning mechanisms to ensure reproducibility and compliance with open-access and FAIR data principles.

LLM-Powered Environmental Q&A System:
- Built a retrieval-augmented generation (RAG) system using LLaMA2 and FAISS over environmental reports.
- Enabled semantic QA over heterogeneous sources including sensor metadata and policy guidelines.

Real-Time Underwater Debris Detection with Optimized YOLOv11 Architecture:
- Developed a lightweight and real-time object detection model using a fine-tuned YOLOv11 architecture optimized for underwater environments.
- Trained the model on an augmented dataset with 11 debris classes captured under low light, turbidity, and variable terrain conditions.
- Achieved 90% mAP on the test set with an F1-score of 89%, while reducing model size by 30% to support deployment on edge devices.
- Evaluated model robustness across diverse viewing angles and lighting scenarios to improve generalization in practical underwater applications.

Breast Cancer Detection and Cell Segmentation:

- Implemented a UNet-based deep learning model to perform precise cell-level segmentation on H&E-stained histopathology images for breast cancer detection.
- Applied data augmentation techniques including rotation, scaling, and elastic deformation to improve generalization and robustness of the model.
- Evaluated segmentation accuracy using metrics such as Dice coefficient and Intersection over Union (IoU) against expert-annotated ground truth masks.

Consumer Emotion Prediction Using EEG Signals:

- Developed an ensemble learning framework combining CNN and LSTM models to classify consumer emotions from multi-channel EEG signals in a neuromarketing context.
- Designed a preprocessing pipeline for EEG data involving artifact removal, band-pass filtering, and feature extraction across time and frequency domains.
- Achieved high classification accuracy across multiple emotion classes and validated performance using subject-independent cross-validation on benchmark datasets.

Peer–Reviewed Publications

- **Ali Shah, S. M.** Casado-Mansilla, D., López-de-Ipiña, D. (2025). *Low-Cost Image-Based PM_{10} Estimation for Citizen Science Air Quality Monitoring (In Review)*. *Environmental Technology & Innovation*, **24**(19), 96425. SCI-Indexed (Q1; IF 7.1).

- **Ali Shah, S. M.**, Casado-Mansilla, D., López-de-Ipiña, D. (2024). *An Image-Based Sensor System for Low-Cost Airborne Particle Detection in Citizen-Science Air Quality Monitoring*. *Sensors*, **24**(19), 96425. SCI-Indexed (Q2; IF 3.9). DOI: 10.3390/s24196425.

- **Ali Shah, S. M.**, Casado-Mansilla, D., López-de-Ipiña, D., Illueca Fernández, E., Hassani, A. (2024). *A Low-Cost Image Sensor for Particulate Matter Detection to Streamline Citizen Science Campaigns on Air Quality Monitoring*. In *International Multidisciplinary Conference on Computer and Energy Science (SpliTech 2024)*. DOI: `10.1109/SpliTechXXX.2024`.

- Pujante Pérez, A., Illueca Fernández, E., **Ali Shah, S. M.**, Jara Valera, A. J., López-de-Ipiña, D. (2024). *Performance Assessment of Wearable Atmotube Pro Sensor for Air Quality Citizen Science Applications*. In *International Multidisciplinary Conference on Computer and Energy Science (SpliTech 2024)*. DOI: `10.1109/SpliTech.2024`.

- **Ali Shah, S. M.**, Muhammad Usman, S., Khalid, S., *et al.* (2023). *An Ensemble Model for Consumer Emotion Prediction Using EEG Signals for Neuromarketing Applications*. **22**(24), 49744. SCI-Indexed (Q2; IF 3.9). DOI: 10.3390/s22249744.

- Mughal, A. B., Haider, S. M., Khan, R. U., **Ali Shah, S. M.**, Hameed Arain, J. (2025, in press). *FireSense: AI-Based Wildfire and Smoke Detection in Forests*. In *International conference on New Trends in Computing Sciences (ICTCS (2025)*.

- Mughal, A. B., **Shah, S. M.**, Haider, S. M., Jabbar, A. (2025). *A Deep Learning Approach to Person Identification via Gait Analysis in Adverse Covariate Scenarios*. In *Proceedings of the 2025 IEEE 4th International Conference on Computing and Machine Intelligence (ICMI)*, Michigan, USA, April 5–6, 2025. ©2025 IEEE.

- Mughal, A. B., Haider, S. M., Khan, R. U., **Ali Shah, S. M.**, Hameed Arain, J. (2025, in press). *FireSense: AI-Based Wildfire and Smoke Detection in Forests*. In *International conference on New Trends in Computing Sciences (ICTCS (2025)*.

Technical Skills

Programming Languages: Python, SQL, Bash, JavaScript

Machine Learning & Deep Learning: TensorFlow, PyTorch, scikit-learn, XGBoost, LightGBM, OpenCV

API Development & Web Services: FastAPI, Flask, RESTful API design, Swagger, JSON

Data Engineering & Formats: Pandas, NumPy, PySpark, Apache Kafka, Spark Streaming, NetCDF, Zarr

Cloud & DevOps: Docker, Kubernetes, Git, GitHub Actions, Jenkins, AWS (S3, EC2), GCP

MLOps & Workflow Automation: CI/CD, Model Versioning, ML Pipeline Orchestration, Experiment Tracking, Agile Development

References

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